

Scenario 1: Student Academic Performance Dashboard

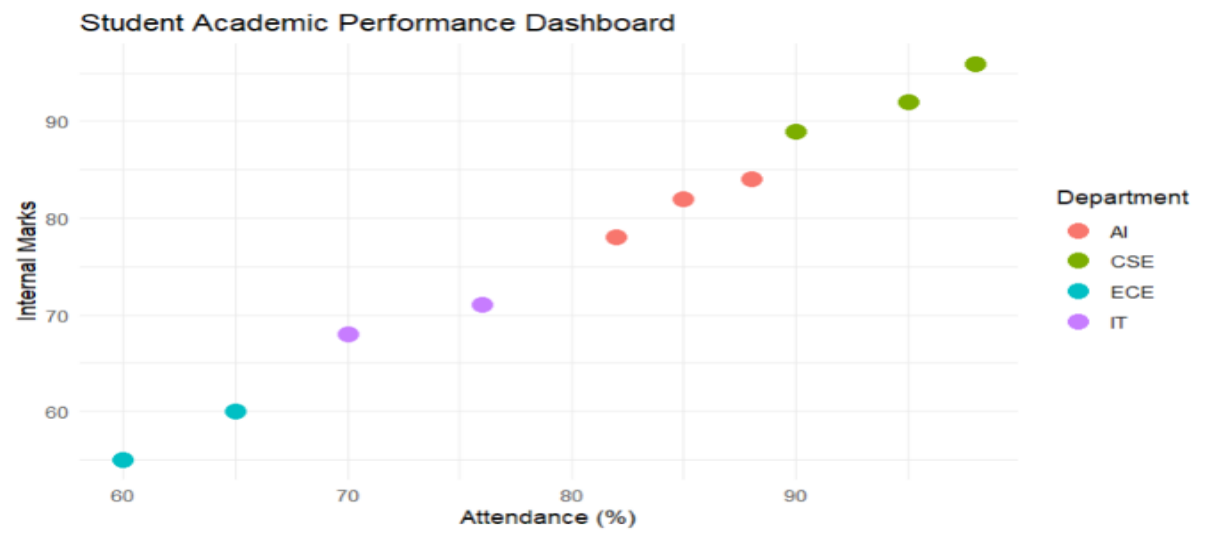
CODING:

```
library(ggplot2)

students <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handeling and
Visualization/students.csv")

ggplot(students,
       aes(x = Attendance,
           y = Internal_Marks,
           color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Academic Performance Dashboard",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```

OUTPUT:



Scenario 2: Electric Vehicle Market Analysis

CODING:

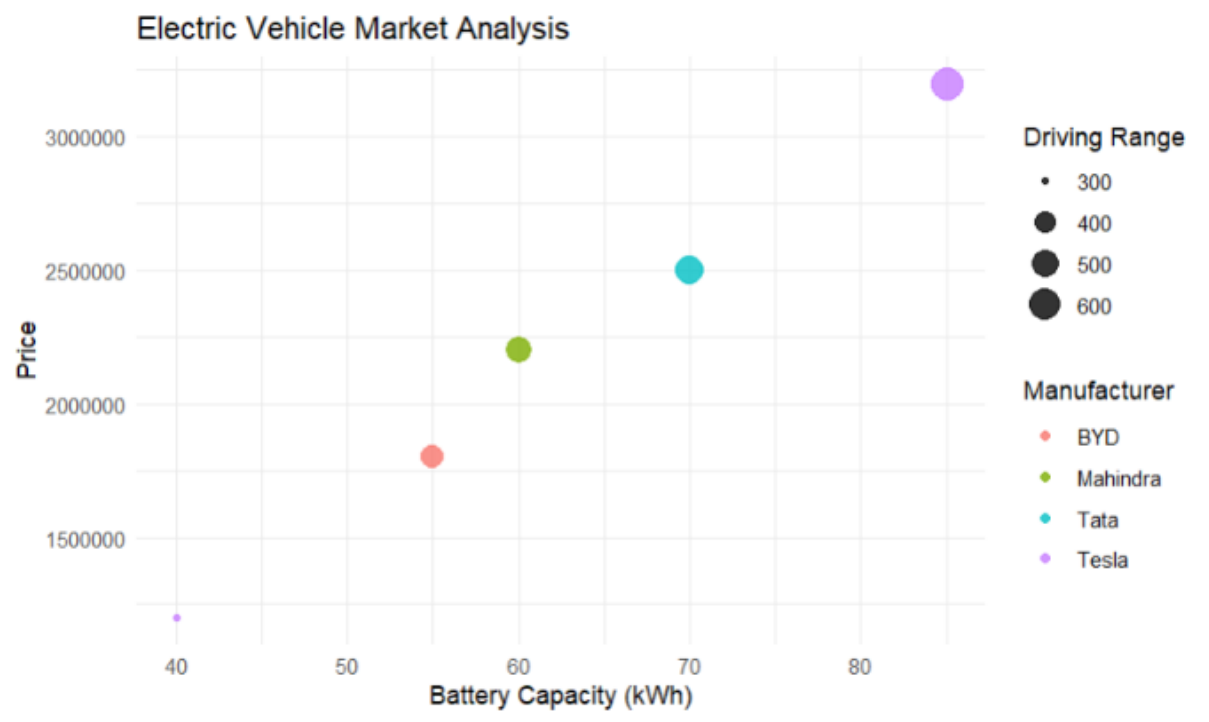
```
library(ggplot2)
```

```
ev <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handling and  
Visualization/ev.csv")
```

```
ggplot(ev,  
  aes(x = Battery_Capacity,  
    y = Price,  
    size = Driving_Range,  
    color = Manufacturer)) +  
geom_point(alpha = 0.8) +  
labs(  
  title = "Electric Vehicle Market Analysis",  
  x = "Battery Capacity (kWh)",  
  y = "Price",  
  size = "Driving Range",  
  color = "Manufacturer"
```

```
) +  
theme_minimal()
```

OUTPUT:



Scenario 3: Hospital Patient Admission Analysis

CODING:

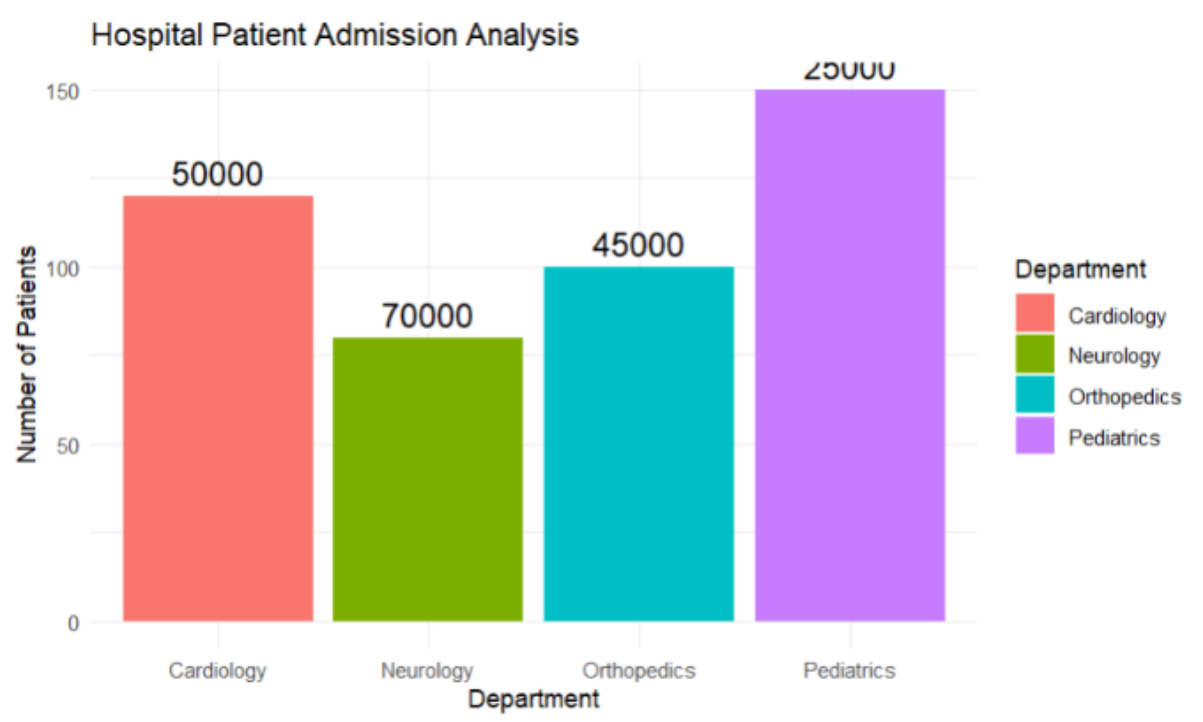
```
library(ggplot2)  
  
hospital <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handling and  
Visualization/hospital.csv")  
  
ggplot(hospital,  
  aes(x = Department,  
    y = Number_of_Patients,  
    fill = Department)) +  
  geom_bar(stat = "identity") +  
  geom_text(aes(label = Average_Treatment_Cost),
```

```

vjust = -0.4,
size = 5) +
labs(
  title = "Hospital Patient Admission Analysis",
  x = "Department",
  y = "Number of Patients"
) +
theme_minimal()

```

OUTPUT:



Scenario 4: Weather Monitoring System

CODING:

```

library(ggplot2)

weather <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handling and
Visualization/weather.csv")

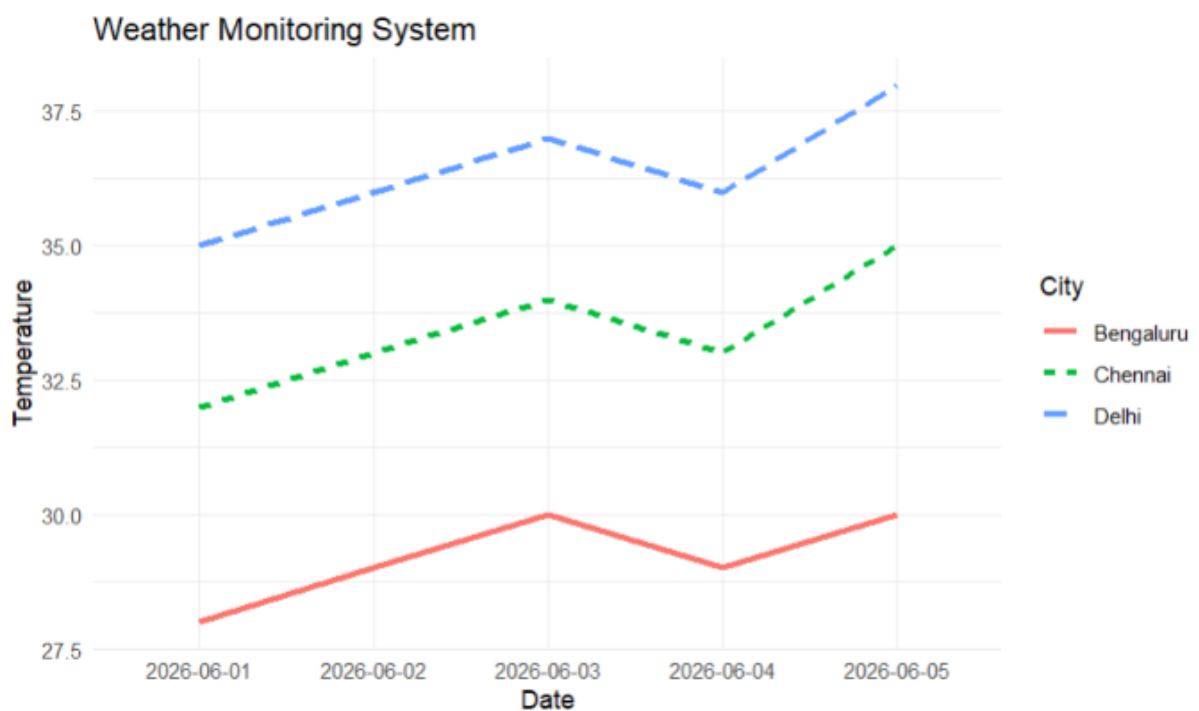
```

```

ggplot(weather,
  aes(x = Date,
    y = Temperature,
    color = City,
    linetype = City,
    group = City)) +
  geom_line(size = 1.2) +
  labs(
    title = "Weather Monitoring System",
    x = "Date",
    y = "Temperature"
  ) +
  theme_minimal()

```

OUTPUT:



Scenario 5: Supermarket Sales Analysis

CODING:

```
library(ggplot2)

sales <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handling and
Visualization/sales.csv")

ggplot(sales,
       aes(x = Product_Category,
           y = Sales_Amount,
           fill = Branch)) +
  geom_bar(stat = "identity",
          position = "dodge") +
  labs(
    title = "Supermarket Sales Analysis",
    x = "Product Category",
    y = "Sales Amount"
  ) +
  theme_minimal()
```

OUTPUT:



Scenario 6: Employee Salary Distribution

```
library(ggplot2)
```

```
employee <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handling and  
Visualization/employee.csv")
```

```
ggplot(employee,  
  aes(x = Salary)) +  
  geom_histogram(  
    binwidth = 5000,  
    fill = "skyblue",  
    color = "black"  
  ) +  
  labs(  
    title = "Employee Salary Distribution",
```

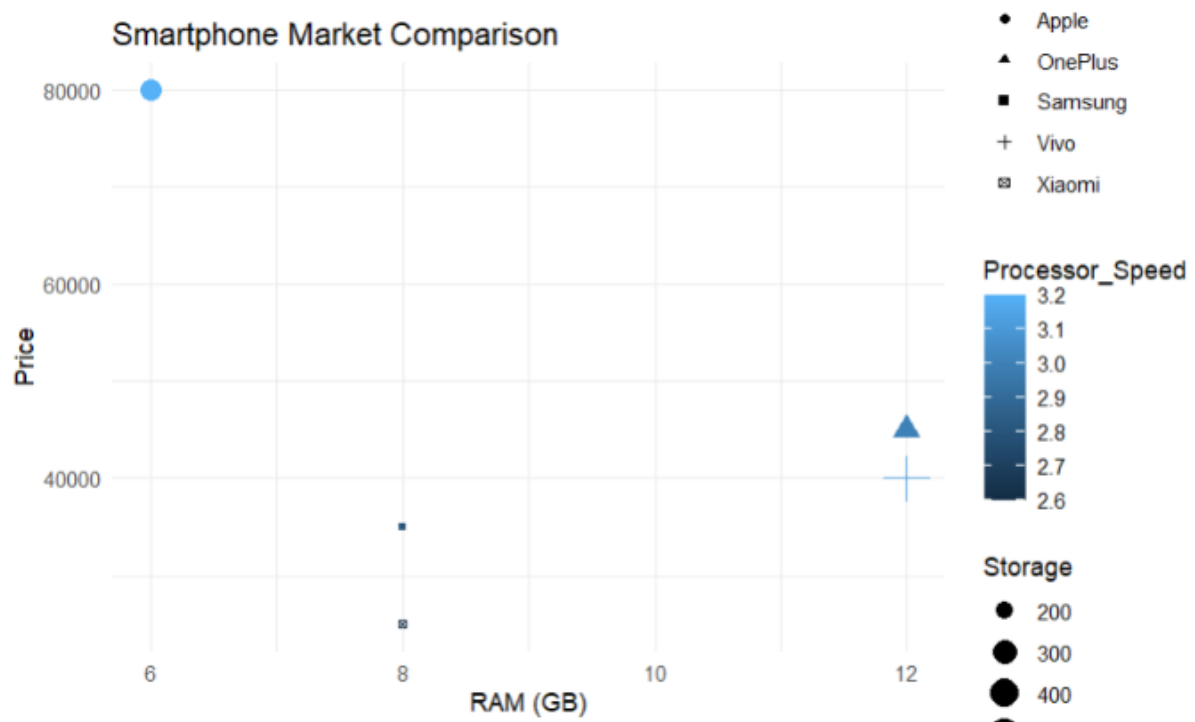
```
x = "Salary",  
y = "Frequency"  
) +  
theme_minimal()
```

Scenario 7: Smartphone Market Comparison

OUTPUT:

```
library(ggplot2)  
  
phone <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handling and  
Visualization/smartphone.csv")  
  
ggplot(phone,  
  aes(x = RAM,  
      y = Price,  
      shape = Brand,  
      size = Storage,  
      color = Processor_Speed)) +  
  geom_point() +  
  labs(  
    title = "Smartphone Market Comparison",  
    x = "RAM (GB)",  
    y = "Price"  
  ) +  
  theme_minimal()
```

OUTPUT:



Scenario 8: Online Movie Rating Analysis

CODING:

```
library(ggplot2)
```

```
movies <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handeling and  
Visualization/movies.csv")
```

```
head(movies)
```

```
ggplot(movies,
```

```
  aes(x = factor(Release_Year),
```

```
      y = Genre,
```

```
      fill = Average_Rating)) +
```

```
geom_tile(color = "white") +
```

```
scale_fill_gradient(
```

```
  low = "yellow",
```

```
  high = "darkgreen"
```

```
) +
```

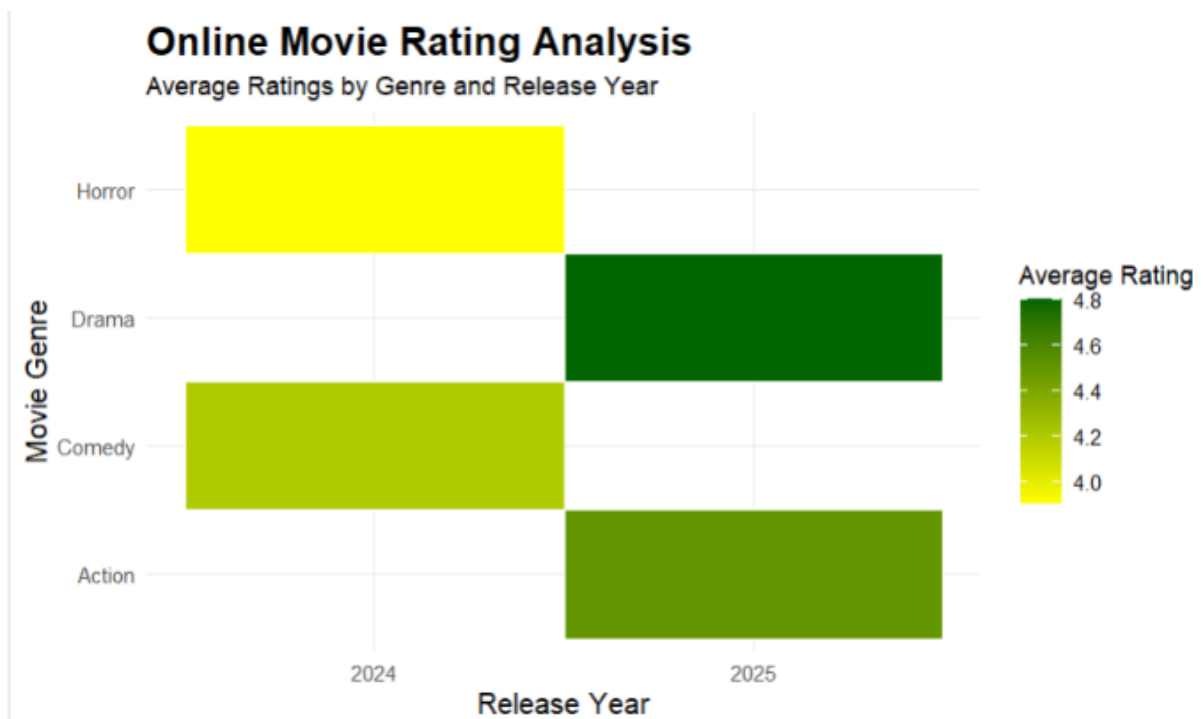
```
labs(
```

```

title = "Online Movie Rating Analysis",
subtitle = "Average Ratings by Genre and Release Year",
x = "Release Year",
y = "Movie Genre",
fill = "Average Rating"
) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold", size = 16),
  axis.title = element_text(size = 12),
  legend.title = element_text(size = 11)
)

```

CODING:



Scenario 9: COVID-19 State-wise Analysis

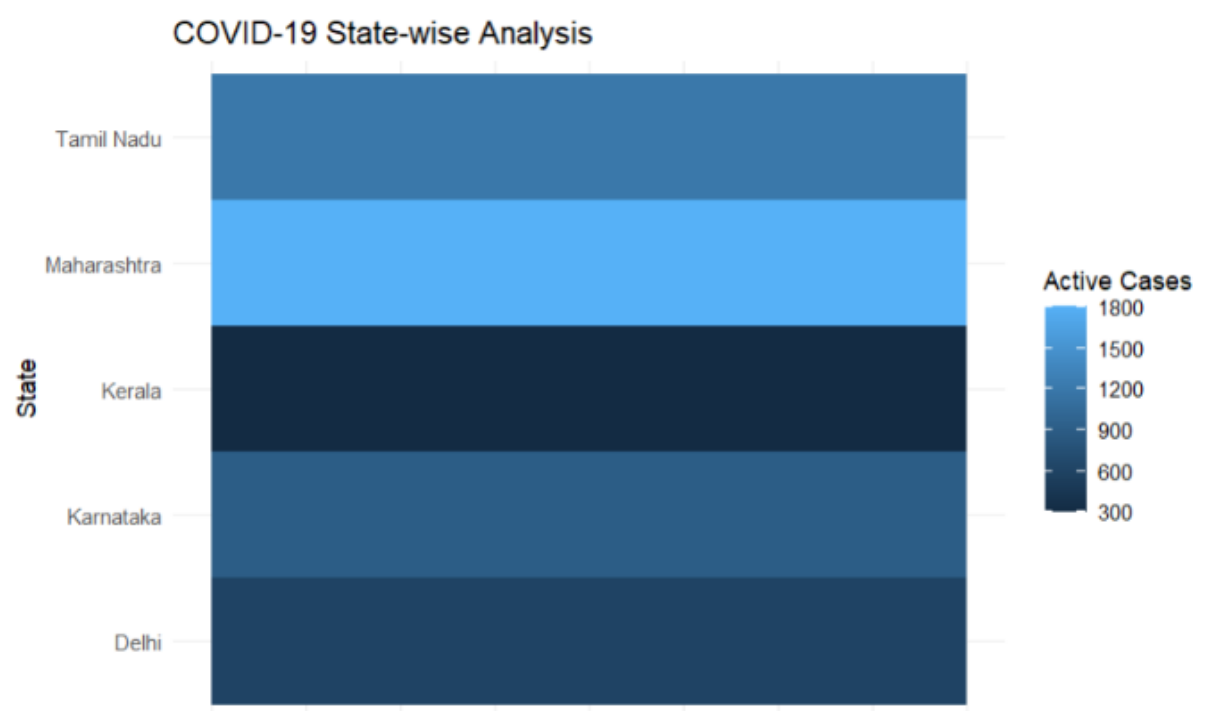
CODING:

```
library(ggplot2)
```

```
covid <- read.csv("C:/Users/senth/OneDrive/Documents/Data Handeling and  
Visualization/covid.csv")
```

```
ggplot(covid,  
  aes(x = 1,  
    y = State,  
    fill = Active_Cases)) +  
geom_tile() +  
labs(  
  title = "COVID-19 State-wise Analysis",  
  y = "State",  
  fill = "Active Cases"  
) +  
theme_minimal() +  
theme(  
  axis.title.x = element_blank(),  
  axis.text.x = element_blank(),  
  axis.ticks.x = element_blank()  
)
```

OUTPUT:



Scenario 10: Iris Flower Classification

CODING:

```
library(ggplot2)
```

```
ggplot(iris,  
  aes(x = Sepal.Length,  
    y = Petal.Length,  
    color = Species,  
    shape = Species,  
    size = Sepal.Width)) +  
geom_point() +  
labs(  
  title = "Iris Flower Classification",  
  x = "Sepal Length",  
  y = "Petal Length"  
) +
```

```
theme_minimal()
```

OUTPUT:

